Local prediction credit

SHoEbox | 13 November 2026 | Recommendation 2 | Base b15baf2

Implemented recommendation 2 on top of neural producer retention: regional local prediction errors now teach their causal child experts, retained-input dependencies and feedback producers. The repair is enabled in fresh reference models and adds no learned parameters: 1,031,952 at 12 experts.

Causal local credit

The saved forecast supplies the issuing expert, activation, input, routing cue and generation-checked producer handles. A synchronous, depth-bounded reverse pass follows the actual context and feedback branches. All neural input and routing derivatives are formed before any expert or router is mutated. Transparent relays preserve credit without spending a neural hop.

Objective and reuse policy

The pass uses the local learning rate, upper-local gain, issuer and recipient local teaching scales, credit gain and the configured reuse exponent. It never enters language/DFA queues. Context and feedback terms for the same consumed role are combined before counting one use. Repeated roles retain their own uses. Retained-input credit follows the old input rather than inventing a gradient through its expert output.

Semantic and freeze behavior

The semantic branch is evaluated once per computed contribution, separately from neural transport, with local teaching scales. Frozen recipients keep their parameters while transmitting derivatives to causal ancestors. A frozen issuer, zero local rate or zero issuer local scale produces no local update.

Checkpoint policy

Version 36 / composition ABI 9 stores the local-credit policy. Transient local work completes inside one teaching call; no additional queued state is serialized. Version-35 checkpoints retain their previous behavior exactly. Explicitly call model.configure_local_credit(True) to upgrade a loaded older model. Historical kernels and explicit historical profiles retain their prior policy.

Verification

175 distinct 1M regression tests passed: the 174-test package run plus the additional live regional test. All 10 local-credit tests also passed with 64 experts, and all seven final reference-policy tests passed. ASan/UBSan passed training, evaluation, drain, reset, serialization and semantic-pool growth.

Independent finite differences check child and feedback derivatives with absolute tolerance 2e-7. Controlled checks cover repeated roles, depth limits, local scales, frozen transmission, untouched language queues, single semantic application, relay transparency, 192 later source executions, and exact save/load. All 12 isolated L3/S3/F3/E4/F4/G5 objectives across two workloads changed causal child experts with language teaching disabled. A real version-35 checkpoint reproduced the previous source version bit-for-bit for parameters, saved composition/neural state and continuation logits.

Measured behavior

Seed 7; one pinned CPU; 2,048 warmup targets followed by 2 rounds of 2,048 targets. Both policies use the same repeated-sentence input, LR 0.04 and local LR 0.04/3. Training-only timing excludes compilation, draining and evaluation. Separate synthetic prose supplies 895 held-out targets. These short runs measure mechanism behavior and throughput.

Measure	Previous policy	Local credit
Median training bytes/s	170.7	129.3
Round 1 training NLL	2.363103	2.361844
Round 2 training NLL	2.071187	2.072541
Synthetic held-out NLL	2.844365	2.844947

Observed median throughput decreased by 24.3%. Wall-clock rates varied across the two rounds (113-228 bytes/s previously, 121-138 with the repair); all individual timings are retained in the raw results. The new run delivered 1,143,087 neural uses, 170,613 feedback branches and 132,875 retained-input uses, with 685,535 descendant neural update calls. All missing local/neural/feedback producers, overwritten credit, expired semantic handles and expired forecasts remained zero.

Frozen expert contribution

Removed transform	Previous delta NLL	Local-credit delta NLL
L3	+0.00054544	+0.00057338
F4	-0.00004384	-0.00004305
G5	+0.00000459	+0.00000289

Each intervention uses a fresh frozen copy of its own trained endpoint. Positive delta NLL means removing that expert transform worsened prediction. Production/diagnostic logits matched exactly, parameters stayed unchanged during evaluation and generation, and chunked continuation was exact for both policies. This smoke does not establish a language-quality improvement.

Generation checks

Temperature 0.8; seed 17; printable ASCII plus newline; prompt: The river

Previous policy: tgh he th he wr th he throe rs wr the the. uChaghe. uhe Thdd heee rne rthee iqe qi th hre there

Local credit: tgh he th he wr th he throe rs wr the the. uEd he the rl#e i!e srde h torrosde The se hhe theie

Reproduce

python -m unittest discover -s experiments/identity_semantics/tests -v

python experiments/identity_semantics/tests/run_completion_sanitizer.py

python experiments/identity_semantics/src/validate_local_credit.py

python experiments/identity_semantics/src/report_local_credit.py

Raw evidence: local_credit_validation.json and local_credit_checks.json. No models, corpus files or binaries are included.